

Finite state automaton 5-tuple

$(Q, \Sigma, \delta, q_0, F)$

- Q is the set of states
- Σ is the alphabet
- δ is the transitions: $Q \times \Sigma \rightarrow Q$
- q_0 is the start state: $q_0 \in Q$
- F is the accepting states: $F \subseteq Q$

DFA Rules

1. Given the current state and symbol we know exactly which state to reach to

NFA Rules

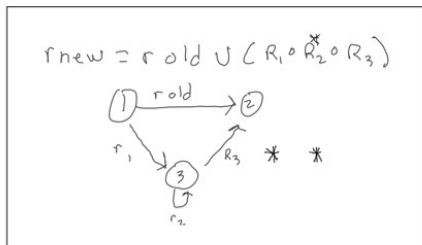
1. Many transition arrows from a state on the same symbol. 
2. Can transition on empty string 

Regular Operations

1. Union: $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$
2. Concatenation: $A \circ B = \{xy \mid x \in A \text{ and } y \in B\}$
3. Star: $A^* = \{x_1 x_2 \dots x_k \mid k \geq 0 \text{ and each } x \in A\}$

$0^* = 0 \text{ or more } 0\text{'s}$

$0^+ = \text{at least 1 } 0 \text{ then } 0 \text{ or more } 0\text{'s} = 00^*$



Regular Expressions

Ex. $(0 \cup 1)0^*$

$= (0 \cup 1) \circ \{0^*\}$

$= (0 \cup 1) \circ \{0^+\}$

- Union ($\cup$)
- Concatenation ($\circ$)
- Kleen star ($*$)

Practice Exam Questions for Regular Expressions

1. $D = \{a^* b \cup b a^*\}$

members: b, ab, aab, ba, baa

2. $E = \{(a \cup b \cup a b a^*)^* (ba)^*\}$

members: $ba, bba, abba$

Pumping Lemma

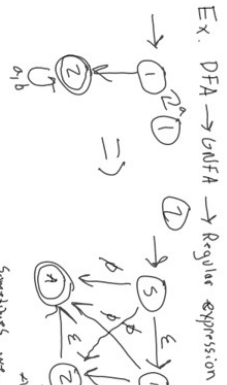
- theorem states that regular languages have a special property
- If we can show that a language does not have this property, then the language is not regular
- Basic Idea of this property is that all strings in a regular language

* If A is a regular language, then there is a number p (the pumping lemma) where if s is any string in A of length at least p , then s may be divided into 3 pieces, $s = xyz$, satisfying the following conditions

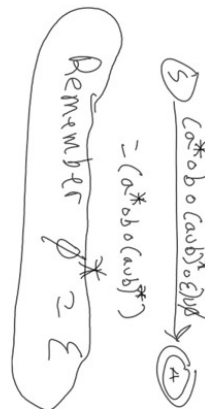
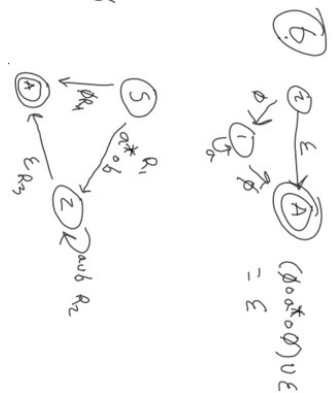
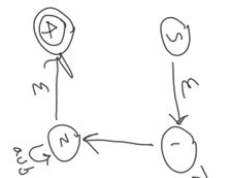
1. for each $i \geq 0$, $x y^i z \in A$
2. $|y| > 0$, and
3. $|xy| \leq p$

• $|s|$ represents the length of the string

• y^i means i copies of string y

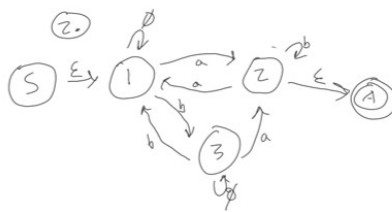
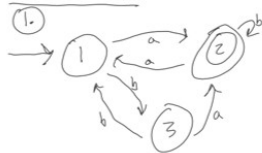


Sometimes we do not include the unifying $\emptyset$ transitions because it clutters the diagram

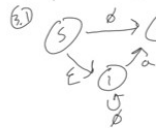


RIPPING EX

Another Example:



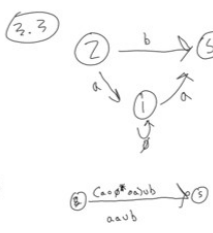
3.1 rip state 1



$$\begin{aligned} 1 & \xrightarrow{(\epsilon \circ \phi \circ a) \cup \phi} 2 \\ &= (\epsilon \circ \phi \circ a) \cup \phi \\ &= a \end{aligned}$$



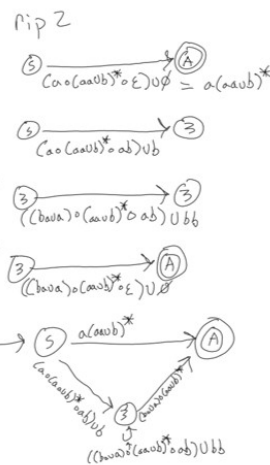
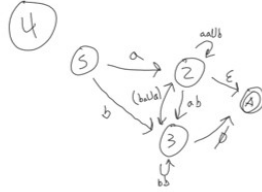
$$\begin{aligned} 5 & \xrightarrow{(\epsilon \circ \phi \circ b) \cup \phi} 3 \\ &= b \end{aligned}$$



$$\begin{aligned} 6 & \xrightarrow{(\epsilon \circ \phi \circ a \circ b) \cup \phi} 5 \\ &= a \circ b \end{aligned}$$



$$\begin{aligned} 3.6 & \xrightarrow{(\epsilon \circ \phi \circ a \circ a) \cup \phi} 2 \\ &= (a \circ a) \end{aligned}$$



$F = \{w \mid w \in \{0,1\}^* \text{ and } w \text{ has an even number of 0's}\}$ show that F is not regular

$101101 \in F$
 $11 \in F$
 $1011 \notin F$

what if we pick $y = 0^p$ as the string y ?
 you cannot use this string for this proof language since y is pumpable.
 $y = 0^p$, let $a \in \mathbb{N}$ such that a is an even #
 how $y = 0^p \circ 0^a \circ 0^p$
 $xy^a z = 0^p \circ 0^a \circ 0^p$
 $|xy^a z| = (p+a) + a + p = 2p + a(1+1)$
 $2p + a(1+1)$ is an even #
 $0^{2p+a(1+1)} \in F$

this is not a good string to pick.

what would a good string be?
 $y = 0^p 1 0^p$

Proof:
 Assume F is regular then it must follow pumping lemma.
 Suppose pumping length is p . let us pick $s = 0^p 1 0^p \in F$
 and $|s| \geq p$ ($|s| = 2p+1$). Then s must be able to be decomposed into $s = xyz$ such that
 1. $|xy| \leq p$
 2. $|y| \geq 1$
 3. $|xy| \leq p$

So by 1) in any decomposition of s , y must only contain 0's, and by 2) y must at least contain one 0. Then suppose $xy^a z = 0^{p+a} 1 0^p$, where $a > 0$ and $p + a(1+1) \leq p$
 $xy^a z = 0^{p+a} 1 0^p \notin F$
 $xy^a z \notin F$ Hence F is not regular

Example: $E = \{0^i 1^j : i > j\}$, $\Sigma = \{0,1\}$ is not regular

$001 \in E$

$01 \notin E$

$10 \notin E$

$S = 0^{p+1} 1^p$

Proof: Assume E is regular then E follows pumping lemma

Pick $s = 0^{p+1} 1^p$, $s \in E$, $|s| = 2p+1 \geq p$

Since E is regular and follows pumping lemma s must be decomposed to $s = xyz$ such that 1. $|xy| \leq p$

2. $|y| \geq 1$

3. $|xy| \leq p$

By 1) and 2) in any decomposition of $s = 0^{p+1} 1^p$, y should only consist of 0's and it should at least consist 1 0. Therefore if you consider $xy^a z$, it should contain either same # of 0's as ones or less # of 0's compared to # of 1's. Hence $xy^a z \notin E$

$s = 0^{p+1} 1^p$, $x = 0^a y = 0^b z = 0^c 1^p$

$a+b \leq p+1$, $b \geq 1$

$xy^a z = 0^{p+1+a} 1^p$ $a \leq p$

$xy^a z \notin E$ $a+b+c = p+1$

$b = p+1 - (a+c)$

$p+1 - (a+c) \geq 1$

$p \geq a+c$

Therefore E is not regular.